

Tim Wang

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Education

University of Sydney (Penultimate Year)

Bachelor of Advanced Computing – Computer Science and Cyber Security

WAM: 78.88 (Distinction) • ATAR: 97.55

Sydney, NSW

Jul 2023 – Expected Nov 2027

Experience

Software Developer | Pancreas Segmentation from Medical Images

Aug 2026 – Oct 2026

Capstone | 7-person team | Client: University of Sydney | Python, PyTorch, nnU-Net, MONAI

- Developing a deep learning model to segment the pancreas from CT and MRI scans, targeting benchmark accuracy against state-of-the-art models.
- Evaluating generalisability across multi-centre datasets for open-source release to the research community.

Software Developer | Refugee English Language Tutoring (RELT)

Aug 2026 – Present

2-person team | SQL, Web Development

- Replacing a manual Excel workflow with a SQL-backed web application for a charity serving 200+ weekly participants.
- Designing an automated matching algorithm to replace manual review, pairing on availability, language, and goals.

Technical Projects

Susquehanna x SYNCS Bot Battle

Annual 2-week competition to develop autonomous bots for a different game environment each year

2026 Competition – 1st of 94 Teams | Python, Optuna, Supabase, PostgreSQL

Jul 2026

Primary developer | 2-person team | Environment based on Agar.io

- Engineered the bot's decision-making system, using game-state tracking to coordinate survival, pursuit, and resource-gathering.
- Built a predictive planning system that evaluated up to 36 movement directions across 14 future time steps, selecting actions by calculating the net present value of projected rewards and risks.
- Developed a distributed optimisation pipeline that ran simulations concurrently across CPU cores on 2 computers, using Optuna with Supabase-hosted PostgreSQL to coordinate trials and accelerate parameter tuning.

2025 Competition – 3rd of 28 Teams | Python

Jul 2025

Primary developer | 4-person team | Environment based on Carcassonne

- Built a bot simulating every tile placement and rotation, scoring structures by ownership and completion likelihood.

2024 Competition – 7th of 72 Teams | Python

Jul 2024

Individual project | Environment based on Risk

- Built a multi-phase decision engine that adapted expansion, attack, and defence to map control and opponent strength.

Lumen Fall | Godot, GDScript

Jan 2024 – Present

2D action game | 2-person team | In development

- Designed a modular, data-driven combat framework supporting 100+ weapon configurations through reusable components.
- Developed a physics-aware A* navigation system that constructs paths from level geometry and validates jumps.

Hivemind | React, Supabase, PostgreSQL

Jun 2026 – Present

Individual project | In development

- Developing a real-time multiplayer web application with room creation, timed rounds, live results, and leaderboards.

Finvolution – Special Category Award | Godot, GDScript

Jun 2026

4-person team | 72-hour USYD Game Jam

- Architected and shipped a browser-playable strategy game at an event involving 200 participants across 44 teams.
- Built a reusable object-oriented framework for 10+ enemy types with configurable spawning, progression, and difficulty scaling.

Leadership Experience

Sydney Computing Society (SYNCS)

Co-President

Sep 2025 – Present

- Co-lead a 54-person executive and subcommittee team serving 5,000+ members, providing direction and oversight across 8 portfolios.
- Manage relations with the School of Computer Science and the Faculty of Engineering, and represent SYNCS in industry partnerships.
- Launched a food sponsorship initiative, established a competition events portfolio, and rebuilt the Notion workspace.

Industry Liaison

Sep 2024 – Sep 2025

- Coordinated events with 16 companies, including a networking night for 160+ attendees.
- Individually secured \$9,000 in sponsorship funding through outreach and negotiations.

Industry Subcommittee

Mar 2024 – Sep 2024

- Organised sponsor events and presented introductory information to 200+ incoming Faculty of Engineering students.

Notion

Campus Leader

Aug 2026 – Present

- Selected as a student ambassador to grow campus adoption through workshops, templates, and product feedback.

Skills

Programming Languages: Python, Java, C, SQL, GDScript, Bash

Tools & Technologies: Supabase, PostgreSQL, Git, Jenkins, JUnit, CI/CD

AI & Optimisation: Hyperparameter Optimisation (Optuna), LLM APIs (OpenRouter), Cursor, Claude, ChatGPT